NAJEEB ABDI

Doha, Qatar | najeebabdi75@gmail.com | (+974) 33047295 | Portfolio: najeeb-abdi.vercel.app | LinkedIn: Najeeb Abdi | GitHub: najeeb101

EDUCATION

B.S. in Computer Science *Expected May 2027*
Qatar University *Doha, Qatar*
Coursework: Data Structures & Algorithms, OOP, Software Engineering, Algorithm Design & Analysis, AI Fundamentals, Machine Learning.

EXPERIENCE

Industry Partnership Intern, Scale AI *Jun 2026 – Aug 2026*
Qatar University Industry Partnership Program *Doha, Qatar*
- Built the CS Intake Planner, an agent-based Monte Carlo simulation of QU CS students moving through the real 41-course study plan (prerequisites, seat limits, pass/fail, dropout) in Python and FastAPI.
- Designed an intake planner that recommends the largest intake within department-set limits (95% uncertainty bounds), with a Next.js dashboard for the Head of Department, a local-LLM advisor (Ollama), and ~286 pytest tests.

AI and Automation Intern *Oct 2025 – Jan 2026*
Silatha (Remote) *Amsterdam, Netherlands*
- Designed and deployed production AI automation pipelines (n8n, Python, OpenAI API) with third-party API integrations, monitoring, and error handling, replacing manual workflows across several business functions.
- Built and deployed a 6-level educational word search game (React, Firebase) with cross-device progress sync, played by ~200 players.

Research Intern, AI-Based Aquaponics System Development *Aug 2025 – Nov 2025*
Qatar University *Doha, Qatar*
- Led AI research on a federated aquaponics control system using reinforcement learning for distributed model optimization, and deployed an AI-enabled greenhouse prototype with ML-driven sensors and control hardware across institutions.

PROJECTS

Thermal Trace, Co-Founder & Lead Developer | Open Innovation Award, UN Reboot the Earth Hackathon *Sep 2026 – Present*
Cold-chain decision platform for Qatar/GCC (FastAPI, MQTT, PostgreSQL, Redis, React) *Doha, Qatar*
- Lead development; built the shelf-life model, an Arrhenius model that turns a load's temperature history into its remaining safe life.
- Enforced a "deterministic math owns every number" architecture: every value carries provenance, and the LLM only explains.

Generative Gains, Founder *Nov 2026 – Present*
Enterprise AI company for Qatar and the GCC; Thermal Trace is its first product. Built the website (Next.js, Tailwind CSS) *Doha, Qatar*

Taqriban, Solo Developer *Oct 2026 – Present*
Arabic-first daily estimation game, live at taqriban.vercel.app *Doha, Qatar*
- Built and launched a mobile-first, right-to-left game (Next.js, Supabase); all data access goes through Postgres security-definer functions with RLS deny-all, so answers stay server-side until a guess is locked.
- Set up CI (Vitest, Playwright, migration guard) and nightly backups; wrote and sourced 40 days of Arabic puzzles.

RouteyAI, Founder & Lead Developer *Mar 2026 – Sep 2026*
School bus route planning and live tracking for Qatar schools *Doha, Qatar*
- Built it solo: K-means clustering plus nearest-neighbour and cheapest-insertion routing on Mapbox driving times, and web and mobile apps for admins, drivers, and parents (Next.js, Expo, Supabase/PostGIS).
- Stopped in Sep 2026 after the Ministry of Education launched its own system covering most of the product.

IEEE Q-Con Conference Website, Sole Developer *Oct 2026 – Present*
- Built the official IEEE Q-Con 2027 site (Astro, GitHub Actions), merging two legacy sites; committee members update it through config, not code.

Robot Swarm Control Hub, ACM QU *Oct 2026 – Present*
- Built the hub for a Raspberry Pi robot swarm (Python, Flask, WebSockets) with live camera feeds, dead-man stops, and per-robot access control.

Other Projects: Lectura, lecture-video-to-notes tool (FastAPI, faster-whisper, map-reduce LLM summaries, Supabase, Docker); AIX Hackathon 2025, bilingual Arabic/English contract analysis prototype in a 3-person team (Streamlit, OpenAI, Azure Translator); Freelance Design & Content (menu design, study guides).

ACTIVITIES & LEADERSHIP

Head of Events, Artificial Intelligence Organization *Apr 2025 – Present*
Qatar University *Doha, Qatar*
- Co-organize campus-wide AI events, including AI Week and the Innovation Program, coordinating logistics and demo setup.

Participant and Ambassador, Shell Intilaaqah Entrepreneurship Program *Sep 2025 – Present*
Shell Qatar *Doha, Qatar*
- Pitched an early-stage startup concept under mentor guidance; hosted (MC) the Fall 2026 masterclass "AI-Powered Innovation".

Also: Member, ACM QU; Events Member, Mechatronics Committee; Student Assistant, IT Student Task Force; Events & IT Support, Research Complex.

SKILLS

AI / ML: Machine Learning, Deep Learning (TensorFlow), Reinforcement Learning, Federated Learning, Monte Carlo Simulation, LLM Apps (OpenAI, Groq, Ollama), n8n
Web & Full-Stack: React, Next.js, TypeScript, Vite, Astro, Expo, FastAPI, Flask, REST APIs, WebSockets, Tailwind CSS
Data & Infrastructure: Supabase (PostgreSQL, PostGIS, Realtime, RLS), Firebase, Redis, MQTT, Docker, GitHub Actions, Vercel, Mapbox GL JS
Languages & Tools: Python, JS/TS, SQL, Java, HTML/CSS, Git, pytest, Vitest, Playwright | **Spoken:** English (Fluent), Arabic